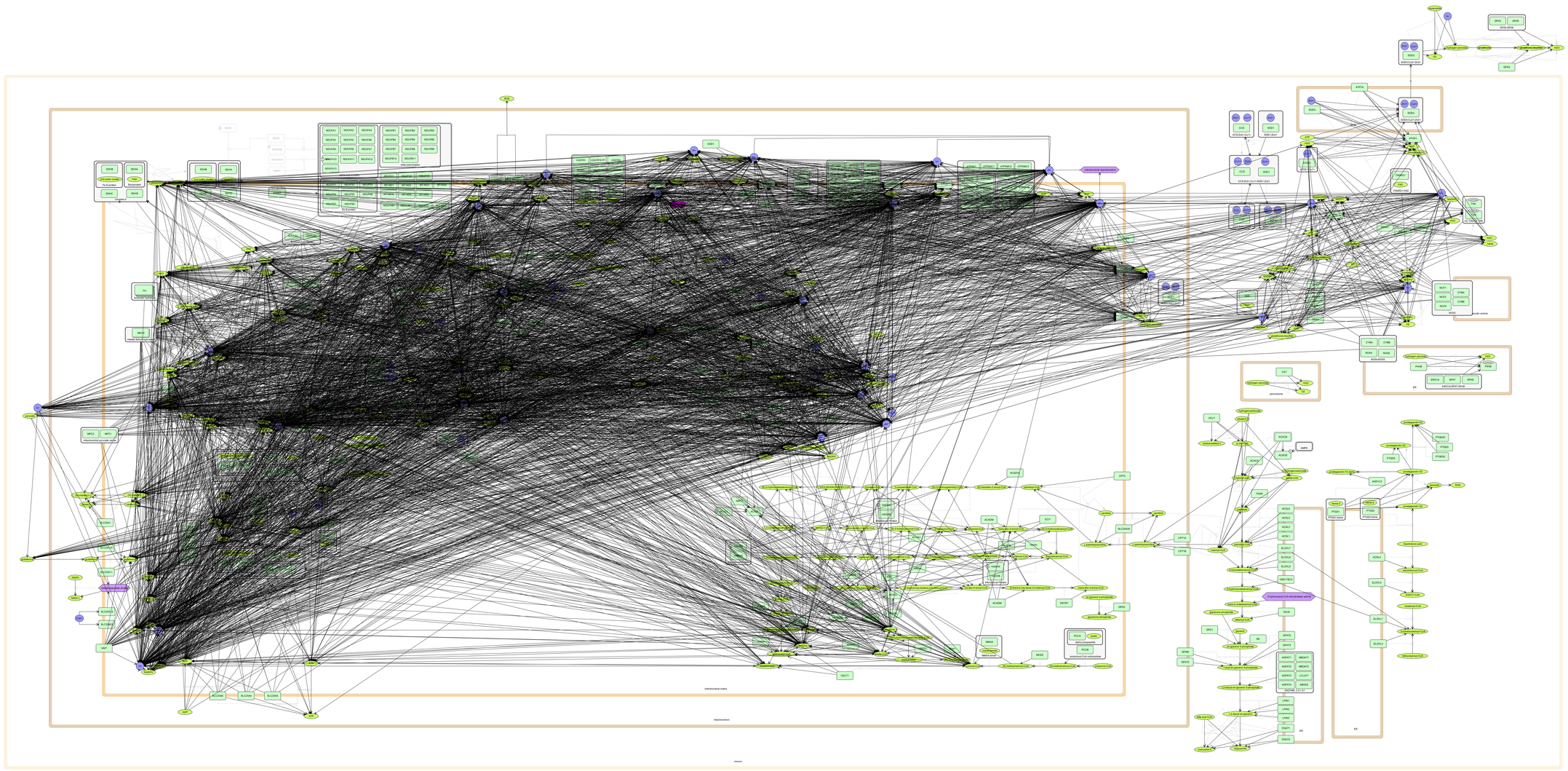

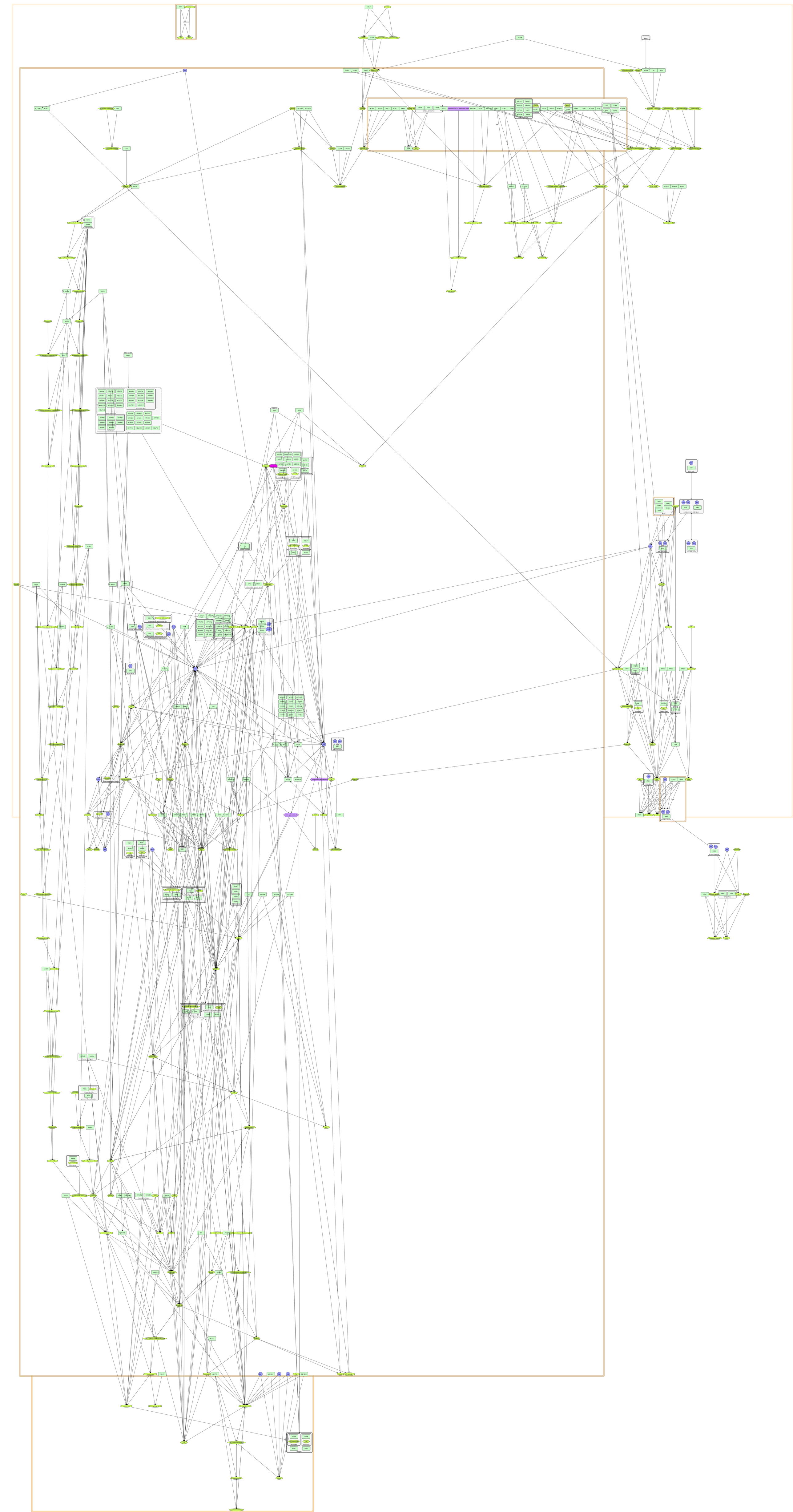

superoxide
H+
Zn2+
Cu2+
SOD3
SOD3:Cu2+:Zn2+
O2
hydrogen peroxide
glutathione
glutathione disulfide
H2O
GPX5
GPX6
GPX5:GPX6
GPX3
ROS
SDHA
SIRT3
NDUFA9
NDUFA10
PINK1
NDUFA1
NDUFA2
NDUFA3
NDUFA4
NDUFA5
NDUFA6
NDUFA8
NDUFA7
NDUFA11
NDUFA12
NDUFA13
alpha subcomplex
NDUFB1
NDUFB2
NDUFB3
NDUFB4
NDUFB5
NDUFB6
NDUFB7
NDUFB8
NDUFB9
NDUFB10
NDUFB11
beta subcomplex
NDUFV1
NDUFV2
NDUFV3
NDUFS1
NDUFS2
NDUFS3
NDUFS4
NDUFS5
NDUFS6
NDUFS7
NDUFS8
Fe-S protein
MT-ND1
MT-ND2
MT-ND3
MT-ND4
MT-ND4L
MT-ND5
MT-ND6
NDUFAB1
NDUFAF1
NDUFC1
NDUFC2
complex I
SDHB
iron-sulfur cluster
Fe-S protein
FAD
flavoprotein
SDHC
SDHD
complex II
FADH2
ubiquinone
PARK7
mutated A53T
SNCA
SUCLG1
SUCLA2
Succinyl-CoA ligase
ATP
fumaric acid
succinic acid
coenzyme A
H2O
FH
fumarate hydratase
(S)-malate(2-)
NAD(+)
MDH2
malate dehydrogenase 2
phosphate(3-)
UQCRH
UQCRFS1P1
UQCRQ
SOD1
CYCS
ATP5IF1
ATP5MC1
ATP5MC2
ATP5MC3
mitochondrial depolarization
ADP
CCS
CCS:Zn2+:Cu1+
Cu1+
SOD1:Zn2+
CCS:Zn2+:Cu1+:SOD1:Zn2+
oxidized
SOD1:Zn2+:Cu2+
ATP7A
Golgi
ATOX1
ATOX1:Cu1+
TXNRD1
TXNRD1:FAD
NADPH
NADP+
TXN
hC TXN:hC TXN
PRDX1
PRDX2
nitrite
peroxynitrite
NO
GSR
GSR:FAD
GPX2
NCF1
NCF2
NCF4
CYBA
CYBB
NOX2
phagocytic vesicle
NOX4
NOX5
NOX4:NOX5
P4HB
ERO1A
GPX7
GPX8
ERO1A:GPX7:GPX8
ER
CAT
peroxisome
pyruvate
MPC2
MPC1
mitochondrial pyruvate carrier
citrate(3-)
SLC25A1
SLC25A10
SLC25A11
NADH
malate-aspartate shuttle
Ca2+
SLC25A12
SLC25A13
NNT
SLC25A6
SLC25A4
SLC25A5
ACADVL
CPT2
(S)-3-hydroxytetradecanoyl-CoA
trans-tetradec-2-enoyl-CoA
myristoyl-CoA
3-oxopalmitoyl-CoA
(S)-3-hydroxypalmitoyl-CoA
(E)-hexadec-2-enoyl-CoA
palmitoyl-CoA
SIRT3
ACADL
HADHA
HADHB
trifunctional Protein
carnitine
3-oxotetradecanoyl-CoA
lauroyl-CoA
trans-dodec-2-enoyl-CoA
(S)-3-hydroxylauroyl-CoA
3-oxolauroyl-CoA
decanoyl-CoA
trans-dec-2-enoyl-CoA
(S)-3-hydroxydecanoyl-CoA
ACADM
ECI1
ECHS1
SLC25A20
L-palmitoylcarnitine
HADH
hexanoyl-CoA
3-oxooctanoyl-CoA
(S)-3-hydroxyoctanoyl-CoA
trans-oct-2-enoyl-CoA
octanoyl-CoA
3-oxodecanoyl-CoA
ACADS
(2-trans,6-cis)-dodeca-2,6-dienoyl-CoA
cis-dec-4-enoyl-CoA
(2-trans,4-cis)-deca-2,4-dienoyl-CoA
trans-dec-3-enoyl-CoA
DECR1
sn-glycerol 3-phosphate
GPD2
glycerone phosphate
HMGCS2
(3S)-3-hydroxy-3-methylglutaryl-CoA
HMGCL
BDH1
ACAT1
acetoacetyl-CoA
acetyl-CoA
acetoacetate
(R)-3-hydroxybutanoate
succinic acid
OXCT1
MMAA
cobamamide
MMAA dimer
MCEE
PCCA
biotin
alpha polypeptide
PCCB
proprionyl-CoA carboxylase
succinyl-CoA
(R)-methylmalonyl-CoA
(S)-methylmalonyl-CoA
propionyl-CoA
mitochondrial matrix
mitochondrion
neuron
ACLY
hydrogencarbonate
oxaloacetate(2-)
ACACB
AMPK
P
ACACA
malonyl-CoA
FASN
palmitate
CPT1A
CPT1B
ACSL6
ACSL5
ACSL3
ACSL1
ELOVL7
ELOVL6
ELOVL3
3-oxooctadecanoyl-CoA
HSD17B12
3-hydroxyoctadecanoyl-CoA
3-hydroxyacyl-CoA dehydratase activity
trans-2-octadecenoyl-CoA
TECR
stearoyl-CoA
GPD1
glycerol
GK
GPAM
GPAT2
GPAT4
GPAT3
1-acyl-sn-glycerol 3-phosphate
AGPAT1
MBOAT1
AGPAT2
MBOAT2
AGPAT3
LCLAT1
AGPAT4
ABHD5
ENZYME: 2.3.1.51
1,2-diacyl-sn-glycerol 3-phosphate
LPIN1
LPIN2
LPIN3
1,2-diacyl-sn-glycerol
fatty acyl-CoA
DGAT1
DGAT2
coenzyme A
triglyceride
prostaglandin E2
PTGES2
PTGES
PTGES3
prostaglandin D2
PTGDS
prostaglandin F2 alpha
AKR1C3
prostaglandin H2
peroxide
heme b
PTGS1
PTGS1:heme
PTGS2
PTGS2:heme
prostaglandin G2
Arachidonic acid
ACSL4
arachidonoyl-CoA
ELOVL5
3ODCT-CoA
icosanoyl-CoA
ELOVL7
3-oxodocosanoyl-CoA
ELOVL4
tetracosanoyl-CoA

CAT
hydrogen peroxide
peroxisome
O2
H2O
ACLY
citrate(3-)
acetyl-CoA
ACACA
hydrogencarbonate
oxaloacetate(2-)
ACACB
AMPK
glycerone phosphate
glycerol
ACACB
GK
GPD1
Ca2+
GPAT2
GPAM
FASN
malonyl-CoA
SLC25A23
AQP8
sn-glycerol 3-phosphate
GPD2
carnitine
SLC25A1
SLC25A20
palmitate
ACSL1
ACSL6
ACSL5
ACSL3
P4HB
hydrogen peroxide
ERO1A
GPX7
GPX8
ERO1A:GPX7:GPX8
TECR
3-hydroxyacyl-CoA dehydratase activity
HSD17B12
ELOVL3
ELOVL6
DGAT2
DGAT1
LPIN2
AGPAT1
MBOAT1
AGPAT2
MBOAT2
AGPAT3
LCLAT1
AGPAT4
ABHD5
ENZYME: 2.3.1.51
heme b
PTGS1
PTGS1:heme
GPAT3
GPAT4
ELOVL7
heme b
PTGS2
PTGS2:heme
LPIN3
LPIN1
ELOVL5
ACSL4
CYBA
CYBB
NOX4
NOX5
NOX4:NOX5
ELOVL4
ER
sn-glycerol 3-phosphate
Arachidonic acid
tetracosanoyl-CoA
icosanoyl-CoA
glycerone phosphate
CPT2
L-palmitoylcarnitine
carnitine
CPT1A
CPT1B
palmitoyl-CoA
P4HB
H2O
1-acyl-sn-glycerol 3-phosphate
prostaglandin G2
arachidonoyl-CoA
3-oxodocosanoyl-CoA
palmitoyl-CoA
ACADVL
L-palmitoylcarnitine
3-oxooctadecanoyl-CoA
AKR1C3
PTGDS
1,2-diacyl-sn-glycerol 3-phosphate
prostaglandin H2
peroxide
3ODCT-CoA
PTGES3
PTGES2
PTGES
3-hydroxyoctadecanoyl-CoA
prostaglandin F2 alpha
prostaglandin D2
fatty acyl-CoA
1,2-diacyl-sn-glycerol
prostaglandin E2
trans-2-octadecenoyl-CoA
triglyceride
coenzyme A
stearoyl-CoA
(E)-hexadec-2-enoyl-CoA
HADHA
HADHB
trifunctional Protein
(S)-3-hydroxypalmitoyl-CoA
ACADL
3-oxopalmitoyl-CoA
SIRT3
linoleoyl-CoA
ACADL
myristoyl-CoA
cis,cis-dodeca-3,6-dienoyl-CoA
ECI1
trans-tetradec-2-enoyl-CoA
PARK7
NDUFA1
NDUFA2
NDUFA3
NDUFA4
NDUFA5
NDUFA6
NDUFA8
NDUFA9
NDUFA7
NDUFA10
NDUFA11
NDUFA12
NDUFA13
alpha subcomplex
NDUFB1
NDUFB2
NDUFB3
NDUFB4
NDUFB5
NDUFB6
NDUFB7
NDUFB8
NDUFB9
NDUFB10
NDUFB11
beta subcomplex
NDUFS1
NDUFS2
NDUFS3
NDUFS4
NDUFS5
NDUFS6
NDUFS7
NDUFS8
Fe-S protein
NDUFV1
NDUFV2
NDUFV3
MT-ND1
MT-ND2
MT-ND3
MT-ND4
MT-ND4L
MT-ND5
MT-ND6
NDUFAB1
NDUFAF1
NDUFC1
NDUFC2
complex I
(2-trans,6-cis)-dodeca-2,6-dienoyl-CoA
(S)-3-hydroxytetradecanoyl-CoA
SNCA
SNCA
cis-dec-4-enoyl-CoA
3-oxotetradecanoyl-CoA
UQCRH
UQCRFS1P1
UQCRQ
BCS1L
UQCR10
UQCR11
UQCRB
UQCRC1
UQCRC2
UQCRFS1
iron-sulfur cluster
iron-sulfur subunit
MT-CYB
heme b
MT-CYB:heme b
complex III
GHITM
CHCHD2
CYCS
HIGD2:GHITM:CYCS
ubiquinol
ROS
lauroyl-CoA
ubiquinone
trans-dodec-2-enoyl-CoA
ECHS1
FH
fumarate hydratase
SDHB
iron-sulfur cluster
Fe-S protein
SDHA
FADH2
flavoprotein
SDHC
SDHD
complex II
pyruvate
HADH
ACADM
(S)-3-hydroxydodecanoyl-CoA
GLUD1
MDH2
malate dehydrogenase 2
MPC2
MPC1
mitochondrial pyruvate carrier
fumaric acid
OGDH
thiamine(1+) diphosphate
2-oxoglutarate decarboxylase (E1)
DBT
lipoamide
dihydrolipoamide succinyl-transferase (E2)
DLD
FAD
dihydrolipoamide dehydrogenase (E3)
alpha-ketoglutarate dehydrogenase
Ca2+
IDH2
Mn2+
isocitrate dehydrogenase
GLUD1
(2-trans,4-cis)-deca-2,4-dienoyl-CoA
DECR1
3-oxododecanoyl-CoA
NNT
ATP5IF1
ATP5MC1
ATP5MC2
ATP5MC3
ATP5PB
ATP5MG
ATP5PD
ATP5MGL
ATP5ME
MT-ATP6
ATP5PF
MT-ATP8
ATP5F1A
ATP5F1B
ATP5F1C
ATP5F1D
ATP5F1E
ATP5F1EP2
complex V
L-glutamate(2-)
(S)-malate(2-)
pyruvate
IDH3A
IDH3B
IDH3G
Mg2+
Mn2+
isocitrate dehydrogenase 3
Mn2+
SOD2
SOD2:Mn2+
CKT
H+
trans-dec-2-enoyl-CoA
decanoyl-CoA
trans-dec-2-enoyl-CoA
rotenone
O2
TXNRD1
TXNRD2
GSR
MT-CO1
MT-CO2
MT-CO3
COX4I1
COX4I2
COX5A
COX5B
COX6A1
COX6A2
COX6B1
COX7A2L
COX7A2
COX7B
COX7C
COX8A
complex IV
mitochondrion
(S)-3-hydroxydecanoyl-CoA
superoxide
NADP(+)
CYCS
SIRT3
CYCS
H+
Zn2+
Cu2+
SOD1
SOD1:Zn2+:Cu2+
3-oxodecanoyl-CoA
Fe2+
paraquat
paraquat monocation radical
hydrogen peroxide
NO
NADPH
SLC25A12
SLC25A11
CYCS
SLC25A12
mitochondrial depolarization
O2
glutathione
octanoyl-CoA
hydroxyl
paraquat
e-
paraquat dication
peroxynitrite
TXN
PRDX3
PRDX1
PRDX5
PRDX2
GPX4
GPX1
glutathione
malate-aspartate shuttle
NADH
superoxide
SOD1
trans-oct-2-enoyl-CoA
ROS
hydroxide
Fe3+
PDP2
PDPR
FAD
PDPR:FAD
PDP2:PDPR
PDP1
PDPR
FAD
PDPR:FAD
PDP1:PDPR
Mg2+
nitrite
TXN
H2O
glutathione disulfide
NAD(+)
hydrogen peroxide
ADP
(S)-3-hydroxyoctanoyl-CoA
thiamine(1+) diphosphate
PDHA1
PDHB
pyruvate dehydrogenase (E1)
DLD
FAD
dihydrolipoamide dehydrogenase (E3)
DLAT
PDHX
pyruvate dehydrogenase complex
PDK1
PDK2
PDK3
PDK4
PDK isoforms
PC
SLC25A6
SLC25A4
SLC25A5
ADP
3-oxooctanoyl-CoA
ACADS
hexanoyl-CoA
NADH
trans-hex-2-enoyl-CoA
thiamine(1+) diphosphate
PDHA1
PDHB
pyruvate dehydrogenase (E1)
DLD
FAD
dihydrolipoamide dehydrogenase (E3)
DLAT
PDHX
pyruvate dehydrogenase complex
(S)-3-hydroxyhexanoyl-CoA
SUCLG1
SUCLA2
Succinyl-CoA ligase
phosphate(3-)
3-oxohexanoyl-CoA
propionyl-CoA
PCCA
biotin
alpha polypeptide
PCCB
propionyl-CoA carboxylase
ATP
butyryl-CoA
(S)-methylmalonyl-CoA
MCEE
oxaloacetate(2-)
ATP
crotonoyl-CoA
MMAA
cobamamide
MMAA dimer
(R)-methylmalonyl-CoA
NAD(+)
OXCT1
(S)-3-hydroxybutanoyl-CoA
succinyl-CoA
ammonium
ACAT1
HMGCS2
SUCLG1
SUCLG2
Succinyl-CoA ligase
GDP
NUDT2
GPR
acetoacetyl-CoA
HMGCS2
GTP
GMP
CS
acetaldehyde
ALDH2
(5-hydroxyindol-3-yl)acetaldehyde
HMGCL
coenzyme A
acetate
ACSS1
(5-hydroxyindol-3-yl)acetic acid
acetyl-CoA
(3S)-3-hydroxy-3-methylglutaryl-CoA
ACO2
citrate(3-)
acetoacetate
BDH1
isocitric acid
ADHFE1
Zn2+
L2HGDH
Mn2+
Co2+
FAD
D2HGDH
citrate(3-)
(S)-malate(2-)
succinic acid
(R)-3-hydroxybutyrate
2-oxoglutaric acid
CO2
(S)-2-hydroxyglutaric acid
SDHB
iron-sulfur cluster
Fe-S protein
SDHA
FAD
flavoprotein
SDHC
SDHD
complex II
4-hydroxybutyrate
FADH2
succinic semialdehyde
Zn2+
SOD1
SOD1:Zn2+
NCF1
NCF2
NCF4
CYBA
CYBB
NOX2
NADPH
Cu1+
Zn2+
Zn2+
CCS
SOD1
CCS:Zn2+:Cu1+:SOD1:Zn2+
H+
Zn2+
Cu2+
SOD1
SOD1:Zn2+:Cu2+
Zn2+
Cu1+
CCS
CCS:Zn2+:Cu1+
O2
superoxide
NO
hydrogen peroxide
GPX1
PRDX6
GSTP1
PRDX6:GSTP1
GPX2
PRDX2
PRDX1
PRDX5
peroxynitrite
glutathione disulfide
GSR
FAD
GSR:FAD
TXNRD1
FAD
TXNRD1:FAD
TXN
TXN
NC:TXN:NC:TXN
glutathione
NADP+
TXN
ATP
Cu1+
ATOX1
ATOX1:Cu1+
H2O
Zn2+
ATP7A
SOD3
nitrite
Golgi
ATOX1
phosphate(3-)
ADP
Zn2+
Cu2+
SOD3
SOD3:Cu2+:Zn2+
Zn2+
Cu2+
SOD3
SOD3:Cu2+:Zn2+
H+
superoxide
GPX3
hydrogen peroxide
GPX5
GPX6
GPX5:GPX6
O2
glutathione
glutathione disulfide
H2O